

Mehraan Chowdhury

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SUMMARY

Machine learning researcher with experience building and evaluating models for computer vision and natural language processing. Comfortable owning a problem end to end, from reviewing the literature and designing experiments, preparing data, adapting open source models, and training them from scratch or fine-tuning pre-trained weights. Works mainly in Python and PyTorch.

Holds an MPhil from UWA, with a thesis ranked in the top 5% and two first-author IEEE publications, and a Master of Information Technology (Artificial Intelligence) from the University of Melbourne with a WAM of 81.1 (H1 average). Also brings 3.5 years of commercial front end development.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher

Sep 2023 - Mar 2026

University of Western Australia, Perth

- Developed novel methods for screening obstructive sleep apnea from 3D facial images, training various machine learning and deep learning models to produce results that outperformed previously published work
- Reviewed the literature for deep learning techniques for classifying 3D meshes and point clouds, identifying high-performing methods that had not been applied to the relevant problem, and designed a program of experiments to test whether they improved classification performance
- Developed a dataset annotation process so medical experts with no technical background could label 3D scans to produce the ground-truth dataset
- Adapted two open-source deep learning codebases, a graph convolutional network and a multi-view CNN, to a new landmarking task, training both models from scratch to locate anatomically significant points on 3D faces
- Built and evaluated a range of machine learning classifiers, applying stratification, cross-validation, normalisation, standardisation, ablation studies, feature reduction, and feature importance analysis to produce results that outperformed previously published work
- Adapted two open-source generative pre-trained point cloud transformers for classification, training under various configurations with and without additional pre-training, both from scratch and from provided pre-trained weights to produce results that outperformed previously published work
- **Tools Used:** Python, PyTorch, scikit-learn, NumPy, Optuna, Open3D, trimesh, MeshLab, CloudCompare, NetworkX, Linux, Git, Cline, Claude Code

Machine Learning Researcher

Jul 2022 - Jan 2023

University of Melbourne, Melbourne

- Built an NLP pipeline in Python to predict relevant programming context from Stack Overflow answers using transformer models (BERT) and NLTK
- Scraped and processed large volumes of developer Q&A data using BeautifulSoup and ChromeDriver

Front-End Web Developer

Apr 2016 - Dec 2019

Revo Interactive Ltd, Dhaka, Bangladesh

- Converted design wireframes into responsive production user interfaces using HTML5, CSS3, JavaScript, and jQuery
- Ensured cross-browser consistency across Chrome and Firefox
- Integrated SEO content into HTML markup to improve search visibility for client websites
- Diagnosed and resolved bugs across the front-end codebase, reducing time-to-fix on reported issues
- Contributed design and development ideas during sprint planning, helping shape feature direction across the team

EDUCATION

Master of Philosophy (Computer Science)

Sep 2023 - Mar 2026

University of Western Australia

- Thesis: *Deep Learning-Based Screening of Obstructive Sleep Apnea Using 3D Facial Images*
- Ranked in the top 5% of theses examined, as noted in the examiner report
- Two first-author IEEE conference publications produced from thesis research

Master of Information Technology (Artificial Intelligence)

Mar 2021 - Dec 2022

University of Melbourne

- WAM: 81.1 (H1 / High Distinction average)
- Thesis: *Predicting Relevant Context for Programming Questions using Stack Overflow Answers*

Bachelor of Science (Neuroscience)

University of Melbourne, Mar 2013 - Nov 2015

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, HTML5, CSS3, SQL, Bash
- **ML & Data:** PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Optuna, Hugging Face, NLTK, Jupyter
- **Computer Vision & 3D:** OpenCV, PIL, Open3D, trimesh, MeshLab, CloudCompare
- **Web automation:** Beautiful Soup, Selenium / ChromeDriver, Playwright
- **AI coding agents:** Claude Code, Cline
- **Tools & Methods:** Git, GitHub, Linux, LaTeX, Agile, Waterfall

PUBLICATIONS

- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Classifying Obstructive Sleep Apnea from 3D Facial Images Using Automated Landmark Detection and Geodesic Distances," *2025 IEEE Conference on Future Machine Learning and Data Science (FMLDS)*, Los Angeles, 2025, pp. 602-607. doi: 10.1109/FMLDS67896.2025.00098
- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Automatic Detection of Peripheral Facial Landmarks Using 3D Facial Data," *2024 IVCNZ Conference on Image and Vision Computing New Zealand*, Christchurch, 2024, pp. 1-6. doi: 10.1109/IVCNZ64857.2024.10794203

REFERENCES

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